ISSN 0015-4628, Fluid Dynamics, 2023, Vol. 58, No. 8, pp. 1662–1667. © Pleiades Publishing, Ltd., 2023.

Review of Numerical and Experimental Works on Studying the Anomalous Ionization Phenomenon

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Received November 24, 2023; revised November 27, 2023; accepted November 27, 2023

Abstract—A brief review of publications on bench experiments focused on investigating the physical processes accompanying the interaction between ionized and neutral gases in the presence of an external magnetic field is presented. The main accent is on the studies concerning the problems of investigation of the anomalous ionization phenomenon when realizing the experiment's parameters (the magnetic and electric field strengths, plasma and neutral gas densities, voltage, and gas type) that are the most convenient for the formulated problem. In addition, the results of the works on numerical simulation of the configurations such as the magnetron, Hall-effect thruster, and Penning discharge are presented in term of interpretation of the phenomena related to the anomalous ionization effect.

Keywords: bench experiments, neutral gas, critical ionization velocity, Penning discharge, magnetron, Hall-effect thruster

DOI: 10.1134/S0015462823602863

INTRODUCTION

The critical ionization velocity effect [1] is observed in the process of relative motion of an ionized gas and a neutral gas in a magnetic field. When increasing the plasma velocity up to some limit (which is called "the critical velocity"), its further growth stops and the energy transferred to the system provides ionization of the particles of the neutral gas. The first hypotheses on the existence of such a phenomenon were formulated by H. Alfvén [1] and confirmed later in a series of laboratory experiments [2–5]. In the reviews [6–8], the possible parameters realized at the setups of different types were systematized; the necessary conditions for manifestations of the anomalous ionization were established; and the mechanisms of emergence of this effect were discussed. In [9], from the generalized experimental data, the efficiency criteria for the enhanced ionization process based on the minimum values of the magnetic field and the neutral-gas density determined in the earlier publications [10] were formulated.

The laboratory investigations performed in the 1970s–1980s were focused on the experimental verification of the Alfvén hypothesis for the following configurations of the setups [6]: plasma guns, unipolar setups with rotating plasma, shock tubes, and collision benches. Moreover, the most important task was to determine not only the fact of manifestation of the anomalous ionization but to justify the mechanism of its emergence. In [8], two main variants of the anomalous ionization of neutral particles were mentioned. The first one is based on the modified two-stream instability of a plasma beam [11], which leads to the turbulent heating of electrons, while the second one is related to the lower hybrid instability, which is a special type of plasma instability and appears due to the relative motion of the ions of the stream and the newly formed secondary plasma [12].

In addition, in this study, the relevant studies [13–22] are considered that concern the problems of revealing the other physical peculiarities accompanying the critical ionization velocity phenomenon and contain the results of the numerical and experimental investigations of the devices in which the gas discharge is applied under the conditions of the presence of a magnetic field: the Penning discharge, Hall-effect thruster, and planar or cylindrical magnetrons (see Fig. 1). Despite the significant differences between the structural peculiarities, as well as the plasma and neutral gas parameters and the magnetic and electric fields, the effects and regularities typical of the anomalous ionization are noted in the devices of such types.

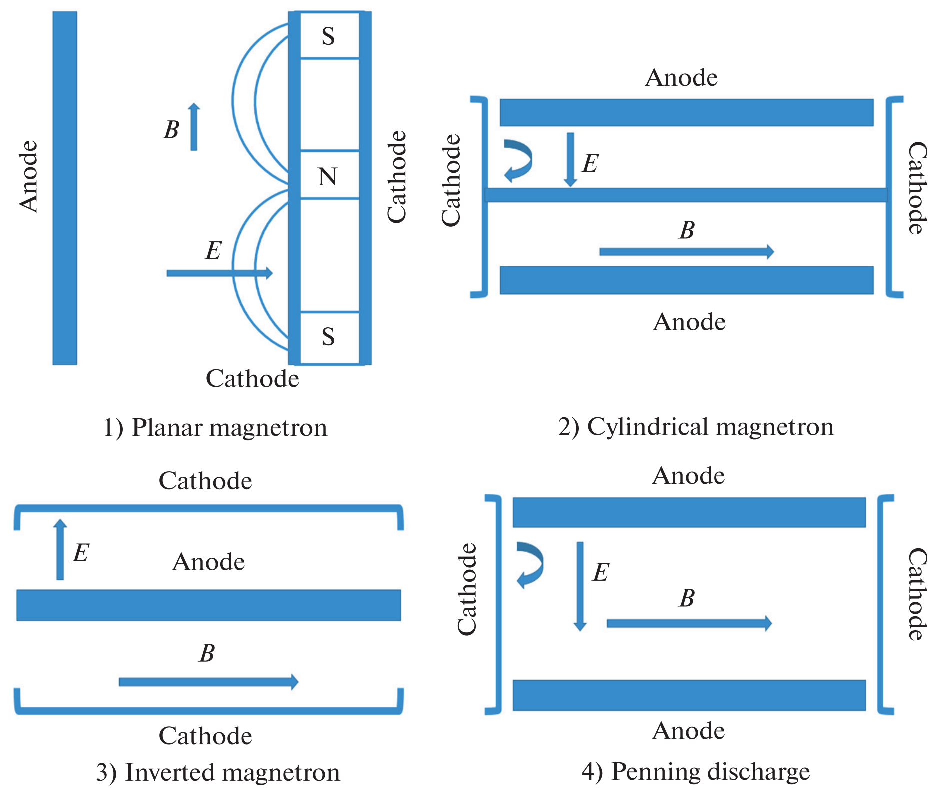

Fig. 1. Comparison of geometrical peculiarities and configurations of the electric and magnetic fields for various devices (from the materials of [21]).

REVIEW OF PUBLICATIONS

In [13], an experimental investigation of a weakly turbulent plasma in the crossed electric and magnetic fields was performed. The bench setup had a structure of the Hall-effect thrust type and included a circular accelerator with an inner radius of 4 cm and an outer radius of 10 cm, an anode made of stainless steel with a slot, a cathode in the shape of a thoriated tungsten filament, a copper coil, and a vacuum chamber with a size of 120 × 140 cm. This configuration provided a radial magnetic and an axial electric field. Argon was used as a neutral gas; it was ionized by the electron impact. The key parameters of a series of experiments are presented in Table 1.

The electron and ion currents, ion velocities and densities, and longitudinal gradient of potential were measured. Ion acceleration due to the action of electrostatic forces was an expected effect; however, at the same time, anomalous electron diffusion in the direction towards the anode was observed; it manifested itself as a much more significant detected value of the electron current in comparison with the analogous value of this parameter conditioned only by collision processes. One of the most important results was

Table 1. Conditions of the bench experiment [13]

Parameter	Value
Axial electric field	~20 V/cm
Average radial magnetic field	~ 475 G
Electron density	10^{10}–10^{11} cm^{-3}
Electron temperature	10–30 eV
Pressure in the chamber	1×10^{-3} Torr
Neutral gas flow rate	0.002 g/s
Final ion velocity	~2–4×10^6 cm/s

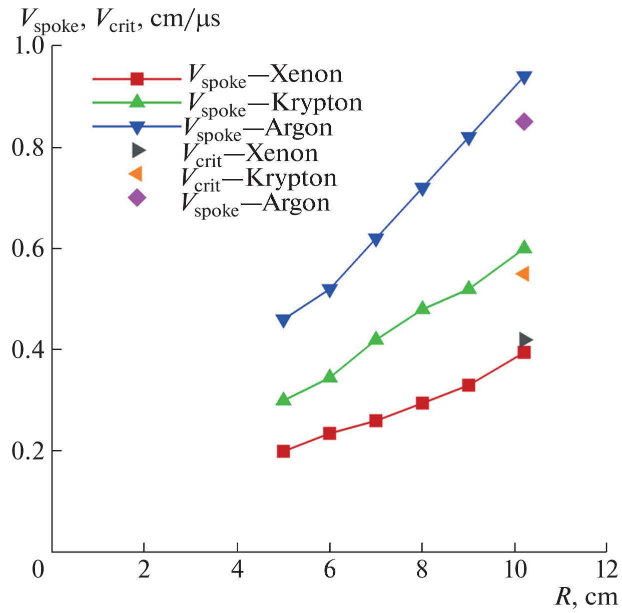

Fig. 2. Comparison of the velocity of a rotating spoke with the values of the critical velocity for argon, krypton, and xenon (according to the materials of [13]). The radial coordinate of the internal wall is $R = 5$ cm; and that of the external wall is $R = 10.2$ cm.

revealing the azimuthal inhomogeneity of density called "the rotating spoke." According to the author of [13], the presence of this structure in the plasma can explain the anomalous diffusion of electrons. In addition, it was noted that the rotating spoke velocity for different inert gases is correlated to the values of the critical ionization velocity (see Fig. 2 [13]) obtained from Alfvén's theory [1]

$$\frac{1}{2}mV_{crit}^2 = eU_{ion}, \tag{1}$$

where m is the mass of neutral-gas particle; U_{ion} is the ionization potential; e is the electron charge; and V_{crit} is the critical velocity. In [13] a hypothesis was proposed suggesting that the effect of anomalous ionization was related to the inhomogeneities in the plasma density.

In [14], experiments were performed to investigate ionization waves. The bench setup was made in the shape of a circular shock tube with a possibility of realization of an azimuthal magnetic displacement field with the value ranging within the interval 3000–5000 G. The glow intensity behind the shock-wave front corresponded to the fully ionized plasma. The key result was determination of the shock wave (its ionization front) velocity, which was quite closely agreed with the critical velocity determined by using formula (1). The data obtained allow discussing the ionization waves as another possible mechanism of anomalous ionization.

In [15, 16], the results of experimental investigations performed with the use of a high-power impulse magnetron sputtering (HiPIMS) cathode, which is an application device using a magnetron discharge and is used, in particular, for deposition of thin films or surface reconstruction. Moreover, this setup has common peculiarities with plasma guns (see, for example, [3]), which were earlier used for the laboratory confirmation of the critical ionization velocity: the value of the magnetic field strength, plasma and neutral gas density, and discharge current [16]. In this investigation performed at a characteristic pressure of ~0.3 Pa and a magnetic field of ~100 mT, the azimuthal inhomogeneity of the plasma was observed and the values of the velocity of the rotating-spoke structure were obtained for various combinations of the neutral gas and the target material close to the theoretical value obtained from formula (1) (see Table 2).

In [17], the results of the numerical investigation of instabilities in the rotating low-temperature plasma in the cylindrical magnetron, in which the mechanism of argon ionization by electron impact was realized, were presented. The argon pressure was chosen from the interval 2–20 mTorr, the magnetic field strength was 10–40 mT, and the applied voltage ranged from 300 to 500 V. The inner radius of the cylinder (cathode) was $R_1 = 0.5$ cm; the outer diameter (anode) was $R_2 = 2.5$ cm. For the cycle of one- and two-dimensional calculations, a combination of the particles-in-cells and Monte Carlo (PIC-MCC) methods

Table 2. Comparison of the theoretical values of the critical ionization velocity V_{crit} and experimental values of the rotating spoke velocity V_{spoke} [15, 16]

Target material	V_{crit}, km/s	V_{spoke}, km/s (krypton)	V_{spoke}, km/s (argon)
Al	6.54	5.4 ± 0.3	8.1 ± 0.3
Cu	4.84	4.2 ± 0.15	4.2 ± 0.4
Nb	3.78	5.7 ± 0.7	7.0 ± 2.0
W	2.89	4.0 ± 0.3	6.25 ± 0.3
Ar	8.72		
Kr	5.68		

was used. The results show that the electric field and plasma density gradient have the same direction. The calculation value of the ion velocity is approximately half the analogous value obtained from the theory of critical ionization velocity. In addition, it is noted that the maintenance of ionization does not require collisionless electron heating.

In [18], the results of the two- and three-dimensional numerical investigation of the Penning discharge in xenon with the use of the PIC-MCC method for the chamber with a size ranging from 5 to 10 cm at a pressure of 1 mTorr and a magnetic field of 100 G are presented. It was suggested to initially inject electrons into the neutral plasma, which provided the ionization and generation of new ions and electrons. The main result obtained was revealing high-frequency rotating-spoke structures. However, as distinct from [13], formation of the rotating spokes was not associated with ionization waves, which did not conform to the anomalous ionization concept. According to the authors of [18], the main mechanism of generation of such structures may be supposedly related to the Simon—Hoh instability. In addition, anomalous electron transport was observed. Moreover, the correlation between the radial electron current and the position of the rotating-spoke structure and, as a consequence, the manifestations of the anomalous electron transport exactly in these regions of the Penning discharge were established.

In [19], the Penning discharge was experimentally investigated on the setup that is a Hall-effect thruster [20]. The discharge-chamber diameter was 2.6 cm; the magnetic field strength varied within 30—500 G; the voltage ranged from 20 to 200 V; and the background pressure was 3 μTorr. The rotating-spoke structures detected in the discharge are highly sensitive to changes in the pressure and magnetic-field strength. It was noted that with growing pressure, the structure was suppressed and, as a consequence, the anomalous electron transport decreased. In contrast, the classical collision mechanism of electron transport becomes dominant.

In [21], rotating plasma structures of three types were investigated: rotating spokes and electron vortices in the cylindrical magnetron, and azimuthal electron-cyclotron drift instabilities in the Hall-effect thruster. At a pressure lower than 10^{-4} Torr, for the modes with a magnetic field of 0.01—0.1 T, it was observed that the plasma lost its quasi-neutrality locally, which occurred because the electron-confinement time was larger than the time of flight of the ions. The calculations performed for the argon discharge in the cylindrical magnetron with a cathode-anode gap of 2—2.5 cm and a pressure of 0.05 mTorr, magnetic field of 50—150 mT, and voltage of 1 kV show the existence of the periodic structures consisting of three electron vortices, which can be subjected to significant changes (vortex spreading) because of the decrease in the electron number in the system. For other conditions (pressure of 10 mTorr, magnetic field of 30 mT, and voltage of 400 V), the characteristic effect that manifests itself in the cylindrical magnetron is formation of a spoke-like structure, which is characterized by violation of the azimuthal symmetry of the distributions of ion and electron density, electric potential, and ion energy. The estimates of the rotational velocity of the spoke structure and the ion velocity presented in [21] give values close to those of the critical ionization velocity.

In [22], the results obtained with the use of the modified diffusion-drift model demonstrate a qualitative possibility of manifestation of the anomalous ionization in the Penning discharge (molecular nitrogen) at a pressure of about 1 mTorr a magnetic field of 0.1 T, and a voltage of 3000 V between electrodes. These conclusions are based on the estimation of the electron energy level that exceeds the potential of impact ionization.

CONCLUSIONS

The most relevant numerical and experimental investigations concerning the phenomenon of critical ionization velocity have been reviewed; it is a continuation of the review that touched only on the issues of the bench and cosmic experiments performed in the 1960s–1990s [23]. In this paper, the laboratory setups realizing various configurations of the electric and magnetic fields at significantly different intervals of pressure, voltage, and plasma density have been considered. The review includes publications on magnetron devices, Hall-effect thrusters, and Penning discharges. The principal possibility of manifestation of the anomalous ionization effect in the devices of all the considered types has been demonstrated. However, it has been noted that, despite the correlation of the calculation and experimental data with the Alfvén theory, the mechanism of the emergence of the anomalously high ionization of neutral particles would be different for the investigated bench setups. Apparently, the most significant difficulties in the reliable determination of the mechanism of emergence of the critical ionization velocity are associated with the Penning discharge.

FUNDING

This study was supported by the Russian Science Foundation, project no. 22-11-00062.

CONFLICT OF INTEREST

The author declares that he has no conflicts of interest.

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Translated by E. Smirnova